

Multimodal Two-Tower Retrieval at Scale: Fusing Product Text and Images for E-Commerce Search

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Abstract

E-commerce customers make purchase decisions by jointly considering product text and visual information, yet most industrial retrieval systems rely almost exclusively on textual signals. Deployed at scale on CPU-based nearest-neighbor infrastructure, these systems leave substantial visual evidence untapped. In this work, we study unified text–image fusion for two-tower retrieval models at a large-scale e-commerce platform, training on 20M query–item pairs. We demonstrate that domain-specific fine-tuning and modality-specific query alignment between user queries and both product text and image modalities are crucial for effective multimodal retrieval. Building on these insights, we propose a mixture-of-modality-experts fusion architecture with a bilinear interaction network to capture cross-modal complementary information. Our model achieves up to 4.86% improvement in nDCG@1 for desirability and 2.36% for relevance over a strong text-only baseline, while preserving the pre-computable item embedding design required for production deployment.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; *Recommender systems*; • **Computing methodologies** → *Multimodal learning*.

Keywords

Multimodal retrieval, Two-tower model, E-commerce search, Image–text fusion

ACM Reference Format:

Qujiaheng Zhang, Guangyue Xu, and Alex Fengjie Li. 2026. Multimodal Two-Tower Retrieval at Scale: Fusing Product Text and Images for E-Commerce Search. In *Proceedings of the Unified Search and Recommendation Workshop (USRW '26)*, co-located with the 20th ACM Conference on Recommender Systems (RecSys '26). ACM, New York, NY, USA, 6 pages. <https://doi.org/XXXXXX.XXXXXXX>

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USRW '26, Minneapolis, MN, co-located with RecSys '26

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ACM ISBN 978-X-XXXX-XXXX-X/YYYY/MM

<https://doi.org/XXXXXX.XXXXXXX>

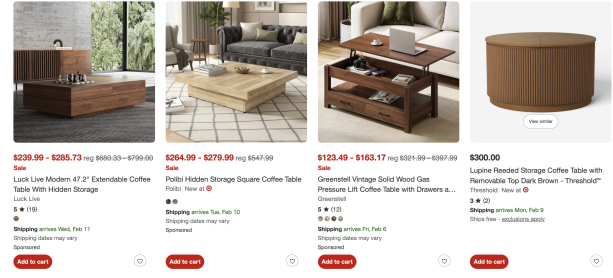


Figure 1: Home decor search is highly visual: product images provide critical relevance signals beyond text, motivating multimodal retrieval.

1 Introduction

Large-scale e-commerce retrieval systems are typically optimized around textual relevance, encoding user queries and product descriptions into a shared embedding space to enable efficient nearest-neighbor search at scale. While this design is computationally attractive and widely adopted in industry [3, 13], it fails to reflect the inherently multimodal nature of user decision-making in online shopping. In practice, as illustrated in Figure 1, users often rely on visual information—such as appearance, style, color, and fine-grained details conveyed by product images—to assess relevance and make interaction decisions, especially when textual descriptions are sparse, ambiguous, or insufficient. This mismatch between the multimodal evidence users consider and the unimodal signals retrieval models are trained on leads to suboptimal relevance modeling, causing text-only retrieval methods to struggle in visually driven categories where subtle visual distinctions dominate relevance judgments. Prior work has shown that incorporating visual signals benefits product retrieval [11, 14] and that multimodal pre-training improves search quality at scale [21].

Despite substantial progress in multimodal representation learning, applying multimodal techniques to e-commerce retrieval poses domain-specific challenges. VL-CLIP [4] points out weak object-level alignment and ambiguous textual representations in product data; EI-CLIP [12] identifies domain-specific entities as confounders and introduces causal interventions; FashionKLIP [16] leverages external knowledge graphs to address the lack of fine-grained cross-modal conceptual knowledge. While these approaches improve multimodal understanding, many introduce additional architectural complexity or computational overhead that makes large-scale nearest-neighbor retrieval challenging [7, 9]. This motivates us to build an efficient multimodal retriever that maintains scalable two-tower search, supports large-scale indexing, and can be deployed in CPU-based production infrastructure.

In this work, we directly address the mismatch between user decision signals and model learning signals in e-commerce retrieval. Our experiments demonstrate consistent improvements of up to 4.86% in nDCG@1 over a strong text-only baseline, confirming that visual signals provide complementary evidence that text alone cannot capture. To achieve these gains, we show that domain-specific fine-tuning is essential for aligning multimodal representations with commercial relevance, and that explicitly aligning user queries with both product text and product images enables the model to learn from the same visual cues that users rely on during relevance judgment. We further propose a lightweight mixture-of-modality-experts fusion architecture combined with bilinear interaction to model fine-grained cross-modal relationships, and introduce a multi-stage curriculum training strategy with a multi-objective loss that jointly models user engagement and semantic relevance.

This work makes the following contributions to large-scale e-commerce retrieval:

- We demonstrate that visual signals provide significant and consistent retrieval improvements, especially for visually driven product categories, and identify domain fine-tuning and modality-specific query alignment as the key enablers.
- We propose a mixture-of-modality-experts fusion architecture with a bilinear interaction network that enables effective integration of textual and visual representations for scalable two-tower retrieval.
- We introduce a multi-stage curriculum training strategy with a multi-objective loss that jointly models user engagement and semantic relevance, and demonstrate consistent improvements across both engagement-based and relevance-based offline evaluation benchmarks.

2 Related Work

Multimodal E-Commerce Retrieval. Industrial retrieval systems have long relied on text-only two-tower models, but growing evidence shows that product images carry complementary relevance signals that text alone cannot capture [11, 14]. Early work on adapting CLIP [15] to e-commerce domains includes VL-CLIP [4], which addresses weak object-level alignment via LLM-augmented embeddings; EI-CLIP [12], which introduces causal interventions for entity-level disambiguation; and FashionKLIP [16], which leverages knowledge graphs for fine-grained cross-modal alignment. CommerceMM [19] and MSRA [8] demonstrate the value of large-scale multimodal pre-training for product search. Amazon’s visual search system [21] shows that multimodal signals improve retrieval in production at scale. Our work differs by focusing on a lightweight, deployable two-tower architecture with systematic ablation of fusion strategies under a curriculum training regime.

Two-Tower Retrieval. Two-tower models [3, 13] are the dominant paradigm for large-scale retrieval due to their efficiency: item embeddings are pre-computed offline and queries are encoded at serving time, enabling sub-100 ms latency at billions of items. Extending this paradigm to multimodal inputs requires careful design to maintain scalability. Fusing modalities at the item tower—rather than at query time—preserves the pre-computation property and keeps serving latency identical to the text-only baseline, which

motivates our fusion-at-item-tower approach. This design also allows the heavier image encoder to run entirely offline, making the system practical for CPU-based production infrastructure.

Curriculum and Multi-Objective Training. Curriculum learning for dense retrieval [6, 10] and hard negative mining [20] have shown consistent gains in text retrieval by progressively increasing training difficulty. We adapt these ideas to the multimodal setting: our three-stage curriculum first adapts encoders to the e-commerce domain, then aligns each modality with user queries independently, and finally trains the full fusion model—a progression that stabilizes optimization and improves final retrieval quality.

3 Proposed Method

3.1 Task Formulation

Given a user query q and a large candidate set of products $\mathcal{X} = \{x_1, \dots, x_n\}$, the goal of e-commerce retrieval is to return the items that best match the query. Depending on the application, matching may reflect semantic relevance between the query and the product, implicit user engagement signals such as clicks and purchases, or a combination of the two. In our setting, each product $x \in \mathcal{X}$ is represented by a textual description such as its title t and a corresponding product image i , which provide complementary semantic and visual evidence for matching estimation.

We aim to learn a function $f(q, x)$ that measures the overall matching degree between a query and a candidate product, and induces a ranking over the candidate set $\mathcal{X}$:

$$x^* = \arg \max_{x \in \mathcal{X}} f(q, x). \quad (1)$$

The optimal item x^* is retrieved via efficient nearest-neighbor search in the shared embedding space.

3.2 Model Architecture

We adopt a standard two-tower retrieval framework for scalable candidate generation in large-scale e-commerce search. The query tower encodes a user query q into an embedding $\mathbf{h}_q$, while the item tower maps each product x into the same semantic space for efficient nearest-neighbor retrieval with cosine similarity.

Query and Item Encoding. We initialize both text and image representations from a pretrained CLIP model [15]. We encode query q via the CLIP text encoder as $\mathbf{h}_q = f_{\text{text}}(q)$, and encode each product’s title t and image i as $\mathbf{h}_t = f_{\text{text}}(t)$ and $\mathbf{h}_i = f_{\text{img}}(i)$.

Mixture-of-Modality Fusion. We integrate complementary semantic and visual signals via a lightweight mixture-of-modality-experts fusion module. A gating network—a two-layer MLP applied to the concatenation of $\mathbf{h}_t$ and $\mathbf{h}_i$ —predicts an adaptive weight $\alpha \in [0, 1]$, and we compute the fused item representation as:

$$\mathbf{h}_f = \alpha \mathbf{h}_t + (1 - \alpha) \mathbf{h}_i. \quad (2)$$

Cross-modal Interaction. To capture fine-grained feature interactions beyond linear fusion, we introduce a multi-head bilinear interaction network [1, 2]. Given the text embedding $\mathbf{h}_t$ and image embedding $\mathbf{h}_i$, we apply K learnable projection heads that combine features via element-wise multiplication, concatenate their outputs across all heads, and pass them through a lightweight MLP. The interaction output is combined with $\mathbf{h}_f$ via a residual connection

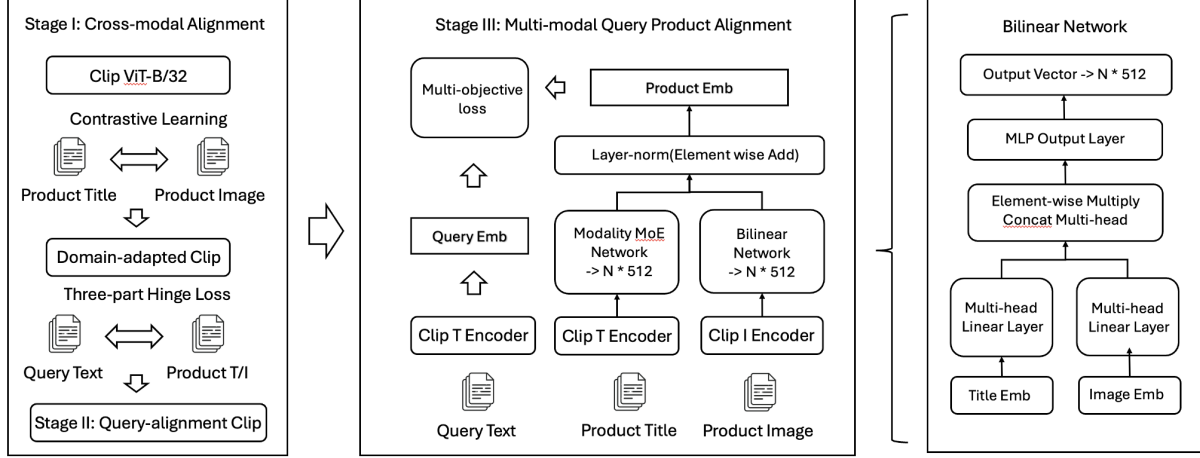


Figure 2: Proposed multimodal retrieval framework. Left: curriculum training procedure for domain adaptation and query alignment; Middle: multimodal two-tower retrieval architecture; Right: bilinear block architecture.

and layer normalization to yield the final item embedding:

$$\mathbf{h}_x = \text{LayerNorm}\left(\mathbf{h}_f + \text{MLP}\left(\left\|\sum_{k=1}^K (\mathbf{W}_t^{(k)} \mathbf{h}_t \odot \mathbf{W}_i^{(k)} \mathbf{h}_i)\right\|\right)\right). \quad (3)$$

Query-item relevance is then measured by cosine similarity:

$$s(q, x) = \cos(\mathbf{h}_q, \mathbf{h}_x). \quad (4)$$

3.3 Self-Adversarial Negative Sampling

We adopt a self-adversarial negative sampling strategy within each mini-batch. Given a batch of N query-product pairs, we compute the cosine similarity matrix between query and product embeddings, where diagonal entries correspond to positives and off-diagonal entries provide candidate negatives. We sample $M = 3$ negatives per query from a softmax-normalized similarity distribution over off-diagonal entries, avoiding the instability of always selecting the hardest negatives while still concentrating sampling probability on more confusable items. As encoders improve during training, the similarity distribution naturally sharpens around harder negatives, providing an automatic curriculum over difficulty [6, 18, 20] without any hyperparameter changes.

3.4 Training Objective

We train the model with two complementary supervision signals: a desirability (engagement) label y_{eng} derived from aggregated user interactions and a semantic relevance label y_{rel} derived from human-aligned cross-encoder scores. Both are graded categorical signals with three levels: high interaction ($y = 2$), low interaction ($y = 1$), and no interaction ($y = 0$).

We adopt a *multi-objective* formulation: unlike a multi-task approach that routes each label to a separate output head, we apply both supervision signals to the *same* predicted score $\hat{y} = s(q, x)$. This design keeps the two-tower architecture intact, preserving pre-computable item embeddings and standard nearest-neighbor retrieval—while allowing the unified embedding space to absorb information from heterogeneous label sources.

Given $\hat{y}$ and a label y , we adopt a three-part squared-hinge loss:

$$\begin{aligned} \mathcal{L}_{3\text{-hinge}}(\hat{y}, y) = & I^+(y) \max(0, \epsilon_+ - \hat{y})^m \\ & + I^-(y) \max(0, \hat{y} - \epsilon_-)^m \\ & + I^0(y) \max(0, \hat{y} - \epsilon_0)^m, \end{aligned} \quad (5)$$

where $I^+(y) = \mathbf{1}[y=2]$, $I^-(y) = \mathbf{1}[y=1]$, and $I^0(y) = \mathbf{1}[y=0]$. The ordered margins $\epsilon_+ > \epsilon_- > \epsilon_0$ encode the desired ranking, leaving an explicit dead zone between adjacent levels so that graded labels are not collapsed into a binary contrast. The exponent m controls how sharply margin violations are penalized. We set $\epsilon_+ = 0.8$, $\epsilon_- = 0.6$, $\epsilon_0 = 0.4$, and $m = 2$, so that large violations are penalized quadratically while small ones contribute mildly.

We combine both label sources in the final objective:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{eng}} \mathbb{E}[\mathcal{L}_{3\text{-hinge}}(\hat{y}, y_{\text{eng}})] + \lambda_{\text{rel}} \mathbb{E}[\mathcal{L}_{3\text{-hinge}}(\hat{y}, y_{\text{rel}})], \quad (6)$$

with $\lambda_{\text{eng}} = 0.7$ and $\lambda_{\text{rel}} = 0.3$, prioritizing engagement as the primary business objective while using relevance as a regularizer that anchors the embedding space to semantic correctness when engagement signals are sparse or noisy.

3.5 Curriculum Training Strategy

We adopt a curriculum-style multi-stage training strategy to progressively align model representations with user decision signals in e-commerce search [6, 10]. We decompose the learning process into three stages with increasing supervision complexity, allowing the model to gradually acquire modality-aware and cross-modal relevance patterns.

Stage I: Domain Adaptation of CLIP Encoders. We fine-tune the CLIP encoders on a large-scale title-image dataset constructed from our e-commerce platform’s product information, using contrastive learning between product titles and product images to adapt pretrained visual-text representations to e-commerce semantics. We adopt the compositional optimization framework of FastCLIP [17], which maintains a moving-average negative sample queue (memory bank) to approximate the full-corpus negative distribution and

enables efficient global contrastive loss optimization without requiring an extremely large batch size.

Stage II: Modality-Specific Query Alignment. We explicitly align user queries with individual item modalities *sequentially*—one epoch per modality—using the three-part hinge loss. This step encourages the model to learn modality-specific relevance signals while incorporating graded ranking supervision, reducing the mismatch between user intent and item representations.

Stage III: Multimodal Fusion Alignment. We align queries with the fused product representations produced by the mixture-of-modality-experts architecture, enabling the model to learn unified multimodal embeddings that capture both unimodal evidence and cross-modal interactions, resulting in more effective retrieval and improved downstream ranking performance.

4 Experiments

We conduct experiments to validate the three key design choices in our method: domain adaptation (Stage I), query alignment (Stage II), and fusion architecture (Stage III). We also evaluate the overall impact of incorporating visual signals and analyze the learned fusion behaviors.

4.1 Training Dataset

We construct our training data from three months of e-commerce search logs, yielding the top 20M aggregated query–item pairs. Each item is represented by its product title and image. We derive a desirability label from aggregated interaction signals (clicks, add-to-cart, purchases) and an auxiliary semantic relevance label from a cross-encoder trained on human-annotated data. The label distribution is approximately 1:3:6 for high, low, and no-interaction labels, reflecting the natural sparsity of strong engagement signals in e-commerce search logs.

4.2 Evaluation Dataset

We evaluate on two held-out benchmarks capturing complementary aspects of retrieval quality. To prevent leakage, all evaluation queries are removed from the training set.

Desirability. A log-based dataset from a disjoint time window reflecting graded user engagement. It contains 248,593 query–item pairs over 6,284 queries, with 40 relevant items per query on average.

Relevance. A human-annotated dataset providing gold-standard semantic judgments. It includes 400,515 pairs from 4,893 queries, with an average of 82 items per query.

4.3 Experimental Setup

We evaluate a series of two-tower retrieval variants to quantify the contribution of visual signals, alignment strategies, and fusion architectures. All multimodal models are built upon CLIP-based text and image encoders, differing only in their adaptation and fusion designs:

- **Text-only (Baseline).** CLIP text encoder over queries and product titles, fine-tuned with the multi-objective loss. This is the text-only baseline used in Table 1.
- **Pretrained CLIP (MoE Fusion).** Multimodal retriever with pretrained CLIP encoders and MoE fusion, jointly fine-tuned

during retrieval training. This serves as the starting point for alignment ablations in Table 2.

- **+FT.** CLIP encoders additionally adapted via Stage I domain fine-tuning on 4M title–image pairs.
- **+FT+Align.** Further adds Stage II modality-specific query alignment.
- **Fusion Ablations.** All fusion variants (MLP, MoE, MoE+MLP, Attention, MoE+Bilinear) share the same +FT+Align encoder initialization and differ only in their fusion module design.

Training Infrastructure. All stages run on two NVIDIA A100 GPUs (batch size 512, AdamW optimizer). Stage I runs for 5 epochs and completes in approximately 4 hours. Stage II runs for 1 epoch per modality, totaling about 3 hours across the two modalities. Stage III runs for 6 epochs and takes roughly 11 hours. The full curriculum thus trains end-to-end in under 18 GPU-hours on a 2×A100 setup, making the framework reproducible within a single-day training budget.

4.4 Metrics

We evaluate retrieval performance using normalized Discounted Cumulative Gain (nDCG@K):

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{r_i} - 1}{\log_2(i + 1)}, \quad \text{nDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}, \quad (7)$$

where r_i denotes the label of the item at rank i and IDCG@K is the DCG of the ideal ranking.

4.5 Experimental Results

We first evaluate the overall impact of incorporating product image information into two-tower retrieval models. Table 1 compares the text-only baseline with the proposed MoE+Bilinear model. The multimodal model yields consistent improvements across all nDCG cutoffs for both desirability and relevance. The gains are largest at the top of the ranking—4.86% at nDCG@1 for desirability and 2.36% at nDCG@1 for relevance—and diminish at deeper cutoffs. This pattern is consistent with the hypothesis that visual signals are most discriminative for distinguishing the single best match from near-miss candidates, where textual descriptions alone are insufficient to resolve fine-grained visual differences. We next ablate the key design choices that contribute to these gains.

4.6 Ablation Studies

Ablation on Domain Fine-Tuning and Query Alignment. We study the effect of different multimodal alignment strategies. Table 2 compares three variants: pretrained CLIP, with domain fine-tuning (+FT), and with additional query alignment (+FT+Align). Domain fine-tuning provides modest but consistent improvements (e.g., +0.1% Desirability@1), indicating better adaptation to e-commerce semantics. Query alignment yields substantially larger gains. The +FT+Align variant improves Desirability@1 by 1.7% over CLIP and Relevance@1 by 0.2%—highlighting the importance of reducing the mismatch between user intent and item representations through explicit modality-level alignment.

Ablation on Fusion Networks. Table 3 compares five multimodal fusion architectures. MoE+Bilinear consistently achieves the strongest performance on both desirability and relevance. Notably,

Table 1: Effect of incorporating product images on retrieval performance (nDCG).

Method	Desirability				Relevance			
	@1	@3	@9	@24	@1	@3	@9	@24
Baseline	0.802	0.809	0.839	0.895	0.890	0.888	0.890	0.897
MoE+Bilinear	0.841	0.839	0.861	0.909	0.911	0.905	0.903	0.909
$\Delta\%$	+4.86	+3.71	+2.62	+1.56	+2.36	+1.91	+1.46	+1.34

Table 2: Ablation on domain fine-tuning and query alignment (nDCG). FT = domain fine-tuning on 4M title-image pairs; Align = modality-specific query alignment (Stage II).

Variant	Desirability				Relevance			
	@1	@3	@9	@24	@1	@3	@9	@24
CLIP	0.824	0.824	0.851	0.903	0.907	0.900	0.899	0.905
+FT	0.825	0.828	0.853	0.904	0.908	0.902	0.899	0.906
+FT+Align	0.838	0.838	0.860	0.908	0.909	0.904	0.902	0.908

Table 3: Comparison of multimodal fusion architectures (nDCG).

Fusion	Desirability				Relevance			
	@1	@3	@9	@24	@1	@3	@9	@24
MLP	0.827	0.829	0.853	0.904	0.908	0.901	0.898	0.905
MoE	0.838	0.838	0.860	0.908	0.909	0.904	0.902	0.908
MoE+MLP	0.836	0.836	0.859	0.908	0.911	0.906	0.902	0.909
Attention	0.833	0.834	0.857	0.906	0.908	0.903	0.900	0.907
MoE+Bilinear	0.841	0.839	0.861	0.909	0.911	0.905	0.903	0.909

MoE+MLP matches MoE+Bilinear on Relevance but falls short on Desirability (0.836 vs. 0.841 at @1): MLP fusion applies a symmetric transformation that cannot capture the asymmetric, multiplicative dependencies between text and image features, whereas bilinear element-wise multiplication naturally models such cross-modal interactions. These results confirm that both adaptive modality weighting and expressive cross-modal interaction are necessary for optimal multimodal retrieval.

Analysis of Learned Fusion Behaviors. To better understand the learned multimodal behaviors, we analyze the fusion weights and interaction outputs of the proposed MoE-based model across product categories. In categories with visually similar products (e.g., apparel), the model assigns higher weight to textual representations where titles are more discriminative; in visually distinctive categories (e.g., consumer electronics), it relies more on image representations. On average, text weight is 0.65, indicating the continued importance of textual information. Bilinear interaction activations are stronger for products where relevance is jointly determined by text and image cues (e.g., furniture with specific styles or functional attributes), confirming that both adaptive modality weighting and explicit cross-modal interaction are essential for effective multimodal retrieval.

4.7 Deployment Considerations

By fusing modalities entirely within the item tower, our architecture ensures that the image encoder and bilinear interaction network

are invoked only during offline embedding computation—never at serving time. This design choice directly enables CPU-only production deployment with zero additional serving overhead compared to the text-only baseline.

The system indexes a corpus of approximately 8 million products, each represented by a 512-dimensional fused embedding stored in a ScaNN [5] approximate nearest-neighbor (ANN) index. At query time, we run only the lightweight CLIP text encoder in the query tower, producing a 512-dimensional query embedding dispatched to the ANN index for top- K candidate retrieval. End-to-end retrieval latency stays consistently under 100 ms (p99), with the system sustaining roughly 2,000 queries per second on commodity CPU-based serving infrastructure. The multimodal model thus deploys as a drop-in replacement without changes to the existing serving stack.

We refresh item embeddings in batch mode: when newly onboarded or updated products exceed a configurable threshold, a batch job re-encodes affected items and atomically swaps the corresponding ANN index shards, amortizing image-encoder cost across millions of items while keeping the online serving path free of the heavier image encoder.

5 Conclusion

Our experiments demonstrate that bridging the multimodal gap in e-commerce retrieval—by incorporating product images into a two-tower framework—yields consistent and substantial improvements in both user engagement and semantic relevance metrics. Domain-specific fine-tuning and explicit query alignment to text and image modalities are the key enablers, and our mixture-of-modality-experts architecture with bilinear interaction provides the most effective fusion strategy among the designs we evaluated. Critically, the fusion-at-item-tower design incurs zero additional serving overhead: product embeddings are pre-computed offline, and the online serving path remains identical to the text-only baseline. These findings provide practical guidance for building scalable multimodal retrieval systems in production. Future work will explore incorporating visual encoders into the query tower to support image-based queries—a direction that requires addressing the latency constraints of online image encoding within the two-tower serving framework.

Acknowledgments

We would like to thank Tianbao Yang and Xiyuan Wei from Texas A&M University for their insightful discussions and valuable guidance during this work. We also acknowledge their open-source FastCLIP codebase, which provided helpful references and implementation insights for our research (<https://github.com/Optimization-AI/FastCLIP/tree/main>).

References

- [1] Hedi Ben-Younes, Rémi Cadene, Nicolas Thome, and Matthieu Cord. 2017. MUTAN: Multimodal Tucker Fusion for Visual Question Answering. In *IEEE International Conference on Computer Vision*. <https://arxiv.org/abs/1705.06676>
- [2] Akira Fukui, Dong Huk Park, Daylen Yang, Anna Rohrbach, Trevor Darrell, and Marcus Rohrbach. 2016. Multimodal Compact Bilinear Pooling for Visual Question Answering and Visual Grounding. In *Advances in Neural Information Processing Systems*. <https://arxiv.org/abs/1606.01847>

- [3] Aditya Gaydhani, Guangyue Xu, Dhanush Kamath, Ankit Singh, and Alex Li. 2026. Unified Learning-to-Rank for Multi-Channel Retrieval in Large-Scale E-Commerce Search. *arXiv:2602.23530 [cs.LG]* doi:10.48550/arXiv.2602.23530
- [4] Ramin Giahi, Kehui Yao, Sriram Kollipara, Kai Zhao, Vahid Mirjalili, Jianpeng Xu, Topojoy Biswas, Evren Korpeoglu, and Kannan Achan. 2025. VI-clip: Enhancing multimodal recommendations via visual grounding and llm-augmented clip embeddings. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems*. 482–491.
- [5] Ruiqi Guo, Philip Sun, Erik Lindgren, Quan Geng, David Simcha, Felix Chern, and Sanjiv Kumar. 2020. Accelerating Large-Scale Inference with Anisotropic Vector Quantization. In *International Conference on Machine Learning (ICML)*.
- [6] Xingwei He, Yeyun Gong, A-Long Jin, Hang Zhang, Anlei Dong, Jian Jiao, Siu Ming Yiu, and Nan Duan. 2023. CAPSTONE: Curriculum Sampling for Dense Retrieval with Document Expansion. In *EMNLP*.
- [7] Mariya Hendriksen, Maurits Bleeker, Svitlana Vakulenko, Nanne Van Noord, Ernst Kuiper, and Maarten De Rijke. 2022. Extending CLIP for Category-to-image Retrieval in E-commerce. In *European Conference on Information Retrieval*. Springer, 289–303.
- [8] Hanqi Jin, Jiwei Tan, Lixin Liu, Lisong Qiu, Shaowei Yao, Xi Chen, and Xiaoyi Zeng. 2023. MSRA: A multi-aspect semantic relevance approach for e-commerce via multimodal pre-training. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*. 3988–3992.
- [9] Zihan Liang, Yufei Ma, ZhiPeng Qian, Huangyu Dai, Zihan Wang, Ben Chen, Chenyi Lei, Yuqing Ding, and Han Li. 2025. UniECS: Unified Multimodal E-Commerce Search Framework with Gated Cross-modal Fusion. *arXiv preprint arXiv:2508.13843* (2025).
- [10] Zhenghao Lin, Yeyun Gong, Xiao Liu, Hang Zhang, Chen Lin, Anlei Dong, Jian Jiao, Jingwen Lu, Daxin Jiang, Rangan Majumder, and Nan Duan. 2023. PROD: Progressive Distillation for Dense Retrieval. In *WWW*.
- [11] Dong Liu and Esther Lopez Ramos. 2025. Multimodal semantic retrieval for product search. In *Companion Proceedings of the ACM Web Conference 2025*. 2170–2175.
- [12] Haoyu Ma, Handong Zhao, Zhe Lin, Ajinkya Kale, Zhangyang Wang, Tong Yu, Jiuxiang Gu, Sunav Choudhary, and Xiaohui Xie. 2022. Ei-clip: Entity-aware interventional contrastive learning for e-commerce cross-modal retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18051–18061.
- [13] Alessandro Magnani, Feng Liu, Suthee Chaidaroon, Sachin Yadav, Praveen Reddy Suram, Ajit Puthenpuhussery, Sijie Chen, Min Xie, Anirudh Kashi, Tony Lee, et al. 2022. Semantic retrieval at Walmart. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 3495–3503.
- [14] Christof Naumzik and Stefan Feuerriegel. 2020. One picture is worth a thousand words? The pricing power of images in e-Commerce. In *Proceedings of The Web Conference 2020*. 3119–3125.
- [15] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. *arXiv preprint arXiv:2103.00020* (2021). <https://arxiv.org/abs/2103.00020>
- [16] Xiaodan Wang, Chengyu Wang, Lei Li, Zhixu Li, Ben Chen, Linbo Jin, Jun Huang, Yanghua Xiao, and Ming Gao. 2023. Fashionclip: Enhancing e-commerce image-text retrieval with fashion multi-modal conceptual knowledge graph. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 5: Industry Track)*. 149–158.
- [17] Xiyuan Wei, Fanjiang Ye, Ori Yonay, Xingyu Chen, Baixi Sun, Dingwen Tao, and Tianbao Yang. 2024. FastCLIP: A Suite of Optimization Techniques to Accelerate CLIP Training with Limited Resources. *arXiv preprint arXiv:2407.01445* (2024).
- [18] Yuqing Xie, Taesik Na, Xiao Xiao, Saurav Manchanda, Young Rao, Zhihong Xu, Guanghua Shu, Esther Vasile, Tejaswi Tenneti, and Haixun Wang. 2022. An Embedding-Based Grocery Search Model at Instacart. In *Proceedings of the ACM SIGIR Workshop on eCommerce (SIGIR eCom)*. ACM, Location TBD. https://sigirecom.github.io/ecom22Papers/paper_8392.pdf arXiv:2209.05555.
- [19] Licheng Yu, Jun Chen, Animesh Sinha, Mengjiao Wang, Yu Chen, Tamara L Berg, and Ning Zhang. 2022. Commercemm: Large-scale commerce multimodal representation learning with omni retrieval. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 4433–4442.
- [20] Kun Zhou, Yeyun Gong, Xiao Liu, Wayne Xin Zhao, Yelong Shen, Anlei Dong, Jingwen Lu, Rangan Majumder, Ji-Rong Wen, Nan Duan, and Weizhu Chen. 2022. SimANS: Simple Ambiguous Negatives Sampling for Dense Text Retrieval. In *EMNLP Industry Track*.
- [21] Xinliang Zhu, Sheng-Wei Huang, Han Ding, Jinyu Yang, Kelvin Chen, Tao Zhou, Tal Neiman, Ouyue Xie, Son Tran, Benjamin Yao, Douglas Gray, Anuj Bindal, and Arnab Dhua. 2024. Bringing Multimodality to Amazon Visual Search System. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*. ACM, 6390–6399. doi:10.1145/3637528.3671640